

Zuhair Khan

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EDUCATION

University of Toronto Scarborough

Sep 2022 – Apr 2027 (Expected)

Honours B.Sc. Computer Science — Software Engineering Specialist (Co-op)

Toronto, ON

- **University of Toronto Scholar Award** — \$7,500 merit scholarship (**top 3%**).
- **Relevant Coursework:** Cybersecurity & Cryptography, Computer Networks, Operating Systems, Systems Programming, Databases, Statistics & Probability, Advanced Data Structures & Algorithms.

EXPERIENCE

Mohawk Medbuy Corporation (MMC)

May 2026 – Aug 2026

Network & Cloud Security Co-op

Burlington, ON

- Investigated **phishing** and **social engineering** incidents using **KnowBe4 PhishER**, **Proofpoint**, and **Cisco Umbrella**, shadowed a third-party **penetration testing** engagement alongside the security team, and resolved **ServiceNow** VPN, MFA, and access control incidents through **Entra ID** and **Exchange** root-cause analysis.
- Administered **IAM** and **RBAC** in the **SAP Ariba** backend across **75+** GTA hospitals, provisioning roles, managing **MFA** lifecycle operations, and reviewing user access for **segregation of duties** conflicts against **ISO/IEC 27001** controls during an **SAP S/4HANA** transformation co-delivered with **Deloitte**.
- Led an **Intune/MDM** compliance initiative retiring stale device records and enforcing endpoint controls via **LAPS-PAM**, and analyzed **Entra ID** exports to flag mailboxes forwarding to **external recipients**, producing a risk-prioritized remediation plan for **Plexus** decommissioning.

University of Toronto Schools (UTS)

Jan 2026 – Apr 2026

IT Systems & Software Engineering Co-op

Toronto, ON

- Built a **SOC-style monitoring dashboard** correlating **Suricata IPS** alerts with **NetFlow**, **RADIUS**, and **NAT** logs across **3 SQL** databases to attribute suspicious network sessions to source IPs and individual users, and troubleshoot **VLAN segmentation** and **DHCP** addressing across academic networks.
- Supported **identity & access management** for **750+** students and staff through **Active Directory** account and group changes, **MFA** enrollment, and **Windows/macOS** endpoint posture under **least privilege**, and helped coordinate a third-party **penetration test** of school systems.
- Contributed to in-house **C#/.NET** web applications using Entity Framework, SQL Server, GraphQL, and Blazor, supporting **Azure DevOps** release workflows through improved **logging** and **PowerShell** automation.

Offline Social Network Inc.

Aug 2025 – Dec 2025

Software & Systems Infrastructure Intern

Toronto, ON

- Led internal **application security risk assessments** for an **AI-driven** mobile app, mapping controls to **OWASP** and **NIST 800-53**, reviewing **authentication** flows (**OAuth 2.0**, **JWT**, bcrypt password hashing), and documenting risk findings while tracking remediation progress to closure.
- Investigated network-layer issues across **AWS** (Route 53, Transit Gateway, Direct Connect), correlating **journald/syslog** output with **tcpdump/Wireshark** traces to accelerate root-cause analysis for **1,000+** users.

SECURITY PROJECTS & RESEARCH | [VIEW ALL](#)

Network Intrusion Detection System (IDS) | [GitHub](#)

Python, PCAP, Detection Engineering

- Built a dual-architecture **NIDS** combining **signature detections** (port scan, SYN/ICMP flood, DNS tunneling, ARP spoofing) with statistical anomaly scoring, validated in an isolated **Docker** testbed using **nmap**, **hping3**, and **arpspoof**.

Web & Application Security CTF Portfolio | [GitHub](#)

Python, OWASP Top 10, Cryptography

- Completed **19 Capture-the-Flag challenges** (UTSC CSCD27) exploiting **OWASP Top 10** classes including **broken access control**, incomplete mediation, **SQL injection**, **XSS**, **CSRF**, and **SSL stripping**, and built **Python** tooling for cryptographic attacks, documenting each with an attack walkthrough and mitigation.
- Practice self-directed **SIEM investigation** exercises on **TryHackMe**, pivoting on failed logins, privilege changes, and network events using **Splunk SPL** to identify suspicious account activity.

Access Control Hardening & Incident Response Labs | [GitHub](#)

Linux, SSH, MFA, Splunk

- Hardened Linux and Mininet VMs with **SSH key-based authentication** and **MFA**, then correlated **PCAP**, authentication logs, and binary artifacts to identify **authentication flaws** and **indicators of compromise** and validate the resulting access-control improvements.

Quantum-Safe Security: PQC & QKD (Undergraduate Research) | [Poster](#)

Python, Qiskit, Cryptography

- Compared **QKD** (BB84, E91) with lattice-based **PQC** (Kyber, Dilithium) under practical **threat models** by building **Python/Qiskit** simulations of noise tolerance and **QBER** thresholds under Prof. Marcelo Ponce, receiving the **2025 CMS Undergraduate Research Symposium award**.

TECHNICAL SKILLS

Security Operations: SIEM (Splunk SPL), IDS/IPS (Suricata), incident investigation, phishing/social engineering analysis, vulnerability assessment, penetration testing (CTF), threat modeling, MITRE ATT&CK, Wireshark, tcpdump, nmap
Security & Risk: ISO/IEC 27001, NIST CSF/800-53, OWASP Top 10, SOC 2, risk assessments, remediation tracking
Identity & Access: IAM, RBAC, MFA, least privilege, SoD, OAuth 2.0, JWT, Active Directory, Entra ID, Intune
Systems & Networks: Windows, macOS, Linux, TCP/IP, VLANs, DNS, DHCP, VPN, firewalls, AWS, Azure, M365, Docker
Tools: ServiceNow, KnowBe4, Proofpoint, Cisco Umbrella, Excel (Power Query), SAP Ariba, Visio, SharePoint
Programming: Python, PowerShell, SQL, Java, C, C++, C#/.NET, TypeScript/JavaScript, Node.js, React, HTML/CSS
Certifications: CompTIA Security+ (in progress), Google Cybersecurity (v2), Cisco Intro to Cybersecurity (2026)